TP03: Interactive Data Analysis and Visualization with Power BI

Big Data Practical Work

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Abstract

This report presents an interactive data analysis and visualization project developed using Microsoft Power BI Desktop. The Amazon Reviews 2018 dataset previously used in TP02 was reused to explore product ratings, review activity, and trends through data transformation, DAX measures, charts, KPI cards, and interactive slicers.

1 Introduction

Data visualization is an important step in the data analysis process because it helps transform large datasets into meaningful information. Microsoft Power BI provides tools for importing, cleaning, analyzing, and visualizing data through interactive reports and dashboards.

The objective of this practical work is to build an interactive dashboard using the Amazon Reviews 2018 dataset previously analyzed in TP02.

2 Objectives

The main objectives of this practical work are:

- Import and explore the dataset in Power BI.
- Clean and transform data using Power Query.
- Create measures using Data Analysis Expressions (DAX).
- Design KPI cards and different chart types.
- Add interactive slicers to filter the report.
- Interpret the results and present the main findings.

3 Dataset and Tools

3.1 Dataset

The project uses the Amazon Reviews 2018 dataset from TP02. The dataset contains product review information, including product identifiers, user identifiers, ratings, and timestamps.

The main columns used in the analysis include:

- `item_id`: identifies a product.
- `user_id`: identifies a reviewer.
- `rating`: represents the product rating.
- `timestamp`: records the review time.

3.2 Tools

The following tools and technologies were used:

- Microsoft Power BI Desktop.
- Power Query for data transformation.
- DAX for creating analytical measures.
- Git and GitHub for project version control.

4 Data Preparation

The dataset was imported into Power BI Desktop and inspected to understand its structure and column types. Power Query was used to prepare the data for analysis.

The preparation process included checking column names, reviewing data types, and examining missing or duplicated values. The relevant columns were organized to support the creation of measures and visualizations.

A review date field derived from the timestamp was also used to support time-based analysis where applicable.

5 Measures and Key Performance Indicators

DAX measures were used to summarize the dataset and provide useful indicators for the dashboard.

The main indicators include:

- Average Product Rating: calculates the average rating across the available reviews.

- Total Ratings: counts the number of ratings included in the analysis.
- Unique Products: counts distinct product identifiers.

These measures provide a concise overview of the dataset and allow users to examine changes when applying filters.

6 Dashboard and Visualizations

The dashboard combines several visual elements to make the data easier to understand.

- **KPI Card:** displays an important summary measure, such as the average product rating.
- **Bar Chart:** compares products or reviewers according to review activity.
- **Column Chart:** compares values across selected categories.
- **Map or Additional Bar Chart:** provides another perspective on the available data.
- **Line Chart:** supports the analysis of review activity over time.
- **Slicers:** allow users to filter the displayed information interactively.

6.1 Final Dashboard

Figure 1 presents the final dashboard created in Power BI Desktop.

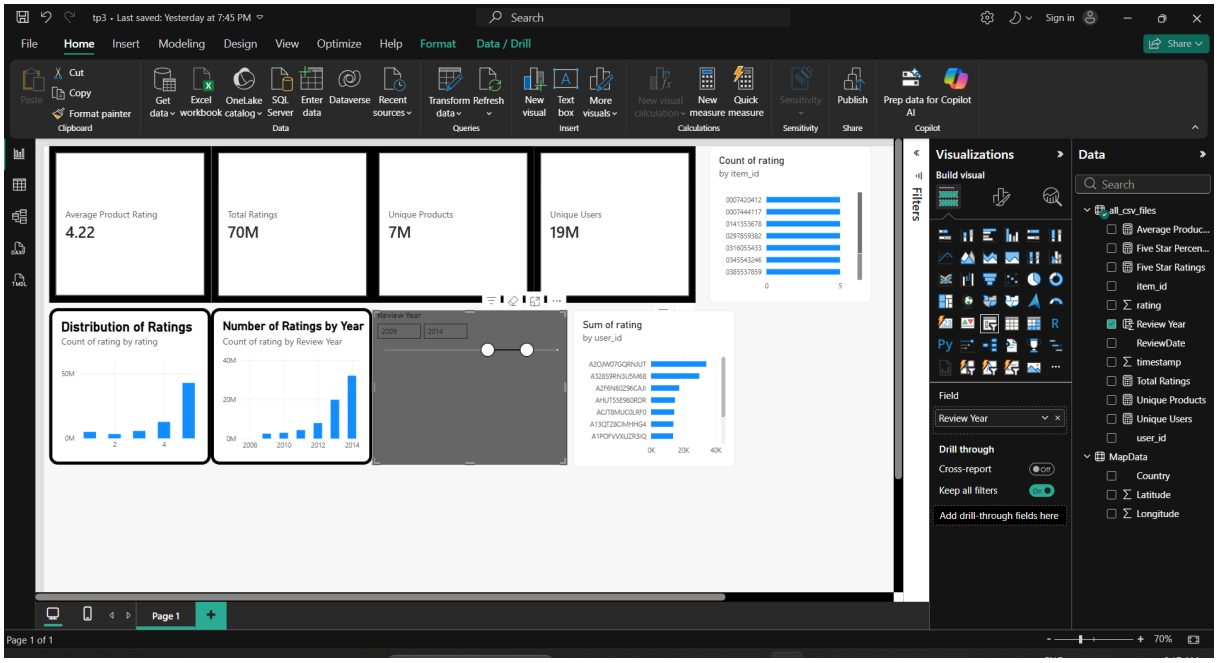


Figure 1: Final interactive Power BI dashboard

7 Results and Findings

The analysis produced the following main findings:

- The average product rating was approximately 4.23.
- The analysis reported approximately 223 million ratings.
- Approximately 15 million unique products were identified.
- KPI cards summarize important information about product ratings and review activity.
- Bar and column charts help compare products and reviewer activity.
- The line chart supports the exploration of review trends over time.
- Interactive slicers allow users to filter the visualizations and explore different subsets of data.

The reported numerical values reflect the results obtained during the previous analysis and should be interpreted according to the final dataset and report filters.

8 Conclusion

This practical work demonstrated the use of Microsoft Power BI for data preparation, analytical calculations, and interactive visualization. Power Query supported data transformation, while DAX measures provided key performance indicators for the dashboard.

The combination of charts, KPI cards, and slicers enables users to explore Amazon review activity from different perspectives. This project also demonstrates how data visualization can support the interpretation and communication of analytical findings.